Ben Guo

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INTERACTIVE PROJECT

Exoplanet Explorer

2025 - Present

space-explorer-bay.vercel.app

Next.js, TypeScript, Three.js, Zustand, Anthropic API

- Built an interactive 3D star map with fly-through navigation across ~4,400 exoplanet systems sourced live from NASA, plus a solar system with 8 planets and 6 moons.
- Implemented first-person movement on planetary surfaces with planet-specific gravity and custom shaders for each planet.
- Rendered thousands of stars in 5 draw calls using instanced meshes grouped by star type and GPU-based particle processing.
- Streamed planet summaries from the Anthropic API, incrementally parsing response fields to display text before generation completed.

EXPERIENCE

Fulgent Genetics

El Monte, CA

Software Engineer

Jan 2025 - Present

- Develop and maintain EzioPath, a digital pathology platform for whole-slide image annotation.
- Eliminated duplicate sample records under concurrent load using atomic database operations and unique constraints.
- Instrumented frontend performance with Datadog RUM and automated Teams alerts to detect load regressions and failed image requests in production.
- Reduced S3 storage costs by 50% through lifecycle rules and cleanup scripts that removed 600K+ stale files.

Software Developer Intern

Jun 2024 - Dec 2024

- Built measurement tools, image filters, and view controls for pathologists working with whole-slide images.
- Accelerated deep zoom and layered tile rendering by 1.7x by redesigning the annotation viewer's image-loading and prefetch pipeline.

EDUCATION

University of California, Los Angeles

Sep 2021 - Dec 2024

B.S. in Computer Science

SKILLS

Languages: Java, Python, TypeScript / JavaScript, C / C++ / C#, SQL, Swift, HTML / CSS

Frameworks: Spring Boot, Node.js, Vue.js, React, Next.js, Three.js, Angular, ASP.NET

Infrastructure: AWS (EKS, S3, IAM, STS, CodeBuild, SES), Kubernetes, Docker, ArgoCD, DocumentDB / MongoDB, Redis

Tooling: Datadog (RUM, monitors, alerting), Git / Bitbucket, CI/CD, Linux